

Unified Power Flow Controller: Operation, Modelling and Applications

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Abstract— This paper focuses on implementing the unified power flow controller (UPFC) in the optimal power flow approach, a critical aspect of power and control system operations. Optimal power flow minimizes operating costs and maintains safety margins for control variables, making it an indispensable tool for energy management. The flexible AC transmission system (FACTS) is a fundamental component of this approach, with the UPFC playing a crucial role. This versatile device provides various types of energy system compensation, enabling the independent control of reactive and active electrical power in transmission lines and bus voltages simultaneously. Previous studies have explored a wide range of engineering applications for UPFC using diverse techniques. This paper reviews these applications, specifically examining how UPFC can increase power system flexibility and controllability. Additionally, this paper explores utilizing artificial intelligence (AI) in the placement of UPFC in power systems. By incorporating AI techniques such as machine learning and optimization algorithms, power system operators can optimize the placement of UPFC to achieve optimal energy management. This approach enhances the efficiency and reliability of energy systems, resulting in significant cost savings and improved power system performance.

Keywords—Power Flow Controller, power flow, transmission system, big data, soft computing, mathematics, mathematical modeling, artificial intelligence, energy consumption, voltage source converter.

I. INTRODUCTION

The expansion of the industrial sector, coupled with the rise in energy consumption and the need to maintain dynamic stability, while ensuring acceptable voltage levels, has resulted in power transmission constraints in the power system [1,2]. To increase the transfer capacity of modern power systems, transmission lines must be built, leading to higher operating costs for these energy systems. Compensators are used to improve the status of existing lines and supply the network with the necessary load [3,4]. High-voltage flexible AC transmission systems are essential for maintaining appropriate voltage levels and quality. Without proper evaluation and accumulation of Flexible AC Transmission Systems (FACTS), the complex power system may be unable to regulate voltage or adjust the level of electrical power injected into or absorbed by the power system. The utilization of FACTS leads to an

overall improvement in grid capacity and performance [5]. Moreover, FACTS devices play a critical role in enhancing the efficiency and reliability of large-scale energy systems. They provide a greater degree of control over electrical energy, enabling the damping of power oscillations. As a result, these devices are instrumental in achieving the flexibility of the power system [6]. A considerable amount of research has been conducted in the field of FACTS devices [7], and they are now a crucial part of interconnected large-scale electrical networks [8]. FACTS devices are classified into three main categories, as illustrated in Fig. 1 [9]. The first category comprises mechanical switches such as thyristor-controlled series compensator (TCSC) [10]. The second category includes hybrid devices such as static synchronous compensator (STATCOM) [11], and the third category includes voltage source converters such as interline power flow controller (IPFC) [12]. By utilizing these FACTS devices, power system operators can efficiently manage power flow and maintain a reliable, stable, and flexible power system.

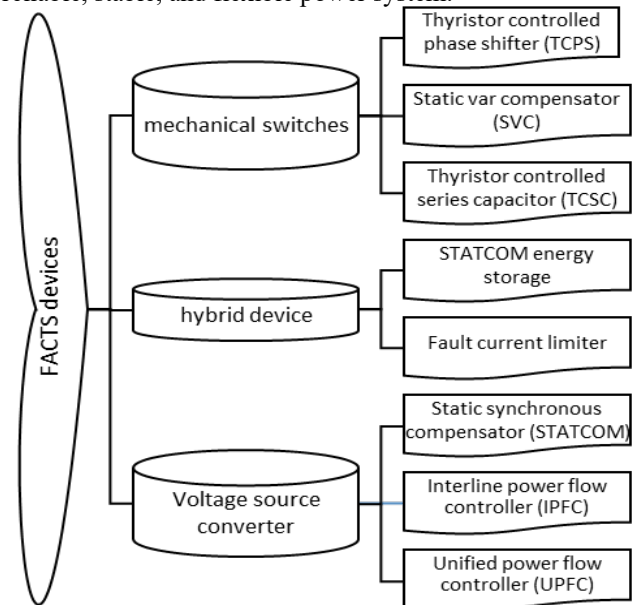


Fig. 1. Classification of FACTS devices

To address the need for reactive power compensation on high-voltage transmission grids, an electrical device called a

Unified Power Flow Controller (UPFC) is utilized [13]. What makes UPFCs unique is their reliance on the protection and control power system, setting them apart from traditional AC transmission technology [14]. Additionally, they are highly adaptable to accommodate the specific requirements of various functionalities, further enhancing their usefulness. In the power system, UPFCs serve different purposes in improving grid behavior, such as security enhancement [15], backup protection [16] and oscillation damping [17]. Due to their numerous advantageous characteristics, UPFCs have been extensively researched for their application in the power system [18,19]. As a member of the FACTS family, UPFCs are connected using a combination of shunt and series connections, making them more flexible in their usage. The aim of this research is to provide a comprehensive review of the various applications of UPFCs in the modern power system. The categorized information presented in Table I summarizes the operating fundamentals of the FACTS device family, including the various possible main control approaches and the local signals utilized for supplementary damping control [20,21]. This concise summary of published research on UPFCs serves as a valuable resource for practitioners and researchers in the field. In conclusion, UPFCs have become an essential component in addressing reactive power compensation in high-voltage transmission grids. Their unique characteristics and adaptability to specific functionalities have made them valuable assets in improving grid behavior.

This paper is structured as follows: Section II of this paper provides a thorough introduction to the principles of unified power flow controllers (UPFC), including their capabilities and limitations. Section III then presents a summary of the coordination design of UPFC and power system stabilizers (PSS), highlighting the key considerations that need to be considered to ensure optimal performance. Moving on to Section IV, this section briefly reviews the current state of research on the application of artificial intelligence (AI) in UPFC optimal placement studies. Specifically, the focus is on how AI can be used to identify the best locations for UPFC installations to mitigate congestion and other related issues. In Section V, this paper categorizes other flexible AC transmission systems (FACTS) devices that are used to address power system challenges, such as sub synchronous resonance (SSR) and congestion. Eventually, this paper is concluded in Section VII.

II. UNIFIED POWER FLOW CONTROLLER

A. Compensating Structure

As illustrated in Fig. 2, the UPFC is a combination of a static synchronous series compensator (SSSC) and a STATCOM that are paired using a prevalent DC voltage source [22]. The UPFC employs a series transformer to inject current into a power transmission line, using a pair of controllable three-phase bridges. This advanced control system allows for the precise management of both reactive and active power flows in the transmission system, providing a balanced sine wave source is applied. One of the key advantages of the UPFC is its ability to independently control the reactive and active power flows on the power line, as well as the bus voltages, thanks to the inverters that operate via a universal DC link and a DC storage capacitor. However, it's worth noting that internal reactive power interaction via a DC link between

two inverters is not possible. The SSSC component of the UPFC is primarily used to regulate the capacity for the transfer of electrical energy in the line to which it is connected, while the STATCOM is typically employed to regulate the bus voltage at locations where typical connections are found [23].

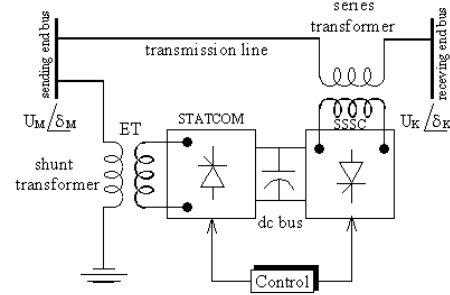


Fig. 2. Orbital structure of unified power flow controller

There are three distinct categories of techniques for controlling the active and reactive power flow: (i) is designed to inject the series voltage in phase shift with the power transmission line current, allowing it to function similarly to a variable sense capacitor [24], (ii) is according to injecting the series voltage in phase shift with the UPFC bus [25], and (iii) the D-Q axis current in the power transferring line is independently controlled, enabling individual control of the reactive and active power flow [26].

TABLE I. Overview of FACTS controller

Type	Devices	Device's principle	Basic control	Local modulation on signal
Serie	TCSC	varying reactance	line compensation	synthesized frequencies synthesized voltages active power, current
	SSSC	reactive source	line compensation	
Shunt	SVC	varying reactance	bus voltage	
	STATCOM	reactive source	bus voltage	
Shunt-series	UPFC	series compensation and reactive source	active power flow compensation of bus voltage line	
Shunt-shunt	BTBL	reactive source and phase compensation	bus voltage active power flow	

B. Compensation Model

The UPFC operates by injecting voltage in series or in shunt with the transmission line. This results in the ability to control both the magnitude and phase angle of the voltage at the point of injection. By controlling these parameters, the UPFC can adjust the flow of power through the transmission line in real-time. This makes it a valuable tool for managing power transmission networks, particularly in situations where the network is under stress or there are variations in demand. The ability to adjust node voltages, line impedances, and phase angle using the UPFC allows for greater flexibility in managing power transmission systems. Additionally, the UPFC can provide voltage support, improve system stability, and mitigate the impact of disturbances on the power network [27]. Since UPFC is a multi-variable power controller in a large-scale energy system, it is essential to investigate the impact of the various power system operating conditions. Fig. 3

illustrates the UPFC model with a regulated voltage source. With the electric grid serving as a representation of the shunt and series voltage source inverters, this model is comprised of two power supplies, one of which is associated with series and the other in shunt. By transforming DC voltage to AC voltage, the power sources are established. The schematic representation of a UPFC with a controlled flow supplier can be found in Fig. 4 [28]. Fig. 5 depicts the model of the UPFC as a transformer with a shunt branch. In this model, the variable shunt susceptance and the turn ratio of the transformer are unaffected by the voltages and currents that are measured at the input and output of UPFC [29]. A dynamic model of UPFC was developed in [30], to boost the power transfer capability through the power transmission network. Series and shunt controllers were structured with fuzzy logic controllers in this approach. In [31], a comparative study on different techniques used to incorporate UPFC in load flow algorithms, such as the decoupled technique, load injection technique, matrix partitioning technique, and indirect technique, is presented. In [32], a reconfigurable cascaded multi-level inverter with a full-bridge converter is suggested. Each phase foot shunt has one end linked to the electrical power line and the other ends linked in parallel to the primary terminals of the series line transmitter and the alternating current (AC) inverter's output terminals.

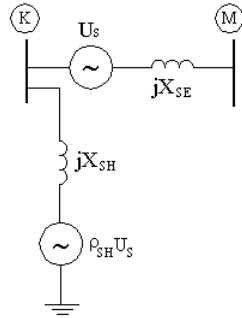


Fig. 3. Equivalent compensation circuit with controlled voltage source

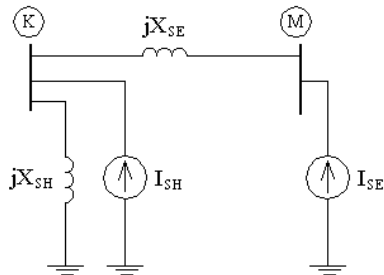


Fig. 4. Equivalent compensation circuit with controlled flow source

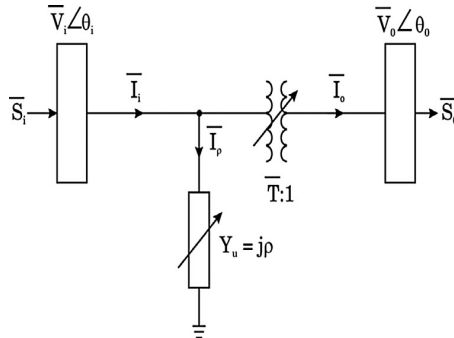


Fig. 5. UPFC Model as a transformer with a shunt branch

III. COORDINATION DESIGN OF UPFC AND PSS

Various techniques have been used for the coordinated design of Power System Stabilizers (PSS) and UPFC controllers. Some of the commonly used techniques include: (i) Classical control design: This applies to developing separate controllers for PSS and UPFC and then combining them using feedback signals. The design is based on linearized models of the power system. (ii) Optimal control design: This involves formulating an optimization problem that aims to minimize a certain cost function while satisfying system constraints. The optimal control design can lead to better performance than the classical control design. (iii) Robust control design: This applies to organizing controllers that can handle uncertainties and disturbances in the power system. Robust control design can improve system stability and performance under varying operating conditions. (iv) adaptive control design: This involves developing controllers that can adapt to changes in the power system. Adaptive control design can improve system performance under changing operating conditions. (v) Artificial intelligence (AI) based control design: This concerns utilizing machine learning and other AI techniques to design controllers for PSS and UPFC. AI-based control design can lead to better performance than traditional control design methods [33,34]. A strategy for coordinating UPFC with PSS to suppress oscillations generated by a small signal disturbance is presented in [35]. This procedure is used to determine the eigenvalue of the greatest real segment and then minimize it as a nonlinear optimization challenge. This model aims to suppress oscillations caused by small signal disturbances. In [36], genetic algorithms are used in a coordinated configuration among a power system stabilizer and a UPFC to optimize the damping proportion of electro-mechanical states by correlating various characteristics of PSSs with a UPFC. An optimal combination for simultaneously locating UPFC and PSS to augment the stability of the power system is addressed in [37], and a mixed integer nonlinear problem is developed for the analysis and design as a result of this presentation research. This architecture aims to boost the transient stability of the large-scale energy system.

IV. USE OF ARTIFICIAL INTELLIGENCE IN UPFC PLACEMENT

Finding the ideal positions and configurations for UPFC devices in power systems is very challenging, and a lot of data is frequently required. Three categories—conventional optimization (CO), sensitivity analysis (SA)-based, and Artificial Intelligence (AI)-based—can be used to categorize the techniques and approaches used in earlier research studies to determine the best locations and settings for the system equipped with UPFC. The most widely used techniques are those based on artificial intelligence, and these are also considered to be the best strategies. Power flow restrictions like reactive and active power, voltage, and power loss are all impacted by the generator failure. The ideal placement of the Facts devices as well as the selection of the appropriate signals in the power system are essential to its effective performance [38]. The best location for the FACTS controllers is extremely important to find the ideal location for UPFC controller placement in multiple applications [39]. Due to its ease of implementation in solving numerous challenging engineering optimization problems, AI is widely used [40]. To prove the practicability of the suggested method for choosing where the

UPFC interface should be in distribution or transmission networks, critical features should be investigated.

The parameters assessed include the phase angle, voltage profile, and proportion of power quality improvement, as well as the cost of the UPFC device during setup and operation, the cost of power generation, the location, number, and parameter of the UPFC device, the deviation of the voltage, the severity index, the voltage stability, and the mitigation of harmonics [41]. These analyses should be carried out in a specific power network, preferably the regular network of the IEEE bus system, under specific contingency conditions. The approach to UPFC placement based on evolutionary programming and various sensitivity analyses is described in [42]. Note that, in the field of optimization, this problem was solved utilizing evolutionary algorithms. To increase dynamic stability, a hybrid strategy based on optimal planning and sizing of UPFC using the combination of the Gravitational Search Algorithm (GSA) and artificial bee colony (ABC) algorithms is developed in [43]. The cuckoo search (CS) and firefly algorithm (FA) are suggested in [44], where the FA strategy optimizes the maximum power loss line as the suitable location of the UPFC, utilizing the best location and the UPFC's capacity to boost the multimachine power system's transient and dynamic stability. To achieve optimal power flow and optimal placement of UPFC, a new gray wolf with a population-based update evaluation algorithm is demonstrated [45]. Furthermore, there are a number of strong approaches, such as [46,47], that can be utilized in the field of planning the location of FACTS devices in the modern power system.

V. CONGESTION MITIGATION AND SSR

There is an extensive power flow approach for the UPFC that is delivered in [48]. This strategy has the capacity to manage both reactive and active electrical powers as well as the voltage amplitude simultaneously. In [49], eigenvalue computation and fast Fourier transform (FFT) investigation against operating point deviations and uncertainties in the system are also analyzed, along with a suggestion for mitigating sub-synchronous resonance (SSR) by employing fractional-order PI (FOPI)-based UPFC. A comprehensive optimization framework according to sequential interpretation to optimally distribute the UPFC and TCSC with wind power generators under deregulated large-scale energy system is furnished in [50], in which the suggested strategy for optimal planning of UPFC and TCSC has been experimented with, and verified on customized IEEE 14-bus and IEEE 118-bus multimachine energy systems.

VI. IMPROVE POWER OSCILLATION DAMPING

Low-frequency oscillations are one of the primary problems that must be solved to guarantee the reliable operation of the power system [51,52]. Power oscillations can be triggered by a variety of factors, including faults in the transmission lines, power line switching, or a sudden change in the output of the generator [53]. Local plant mode oscillations, interplant mode oscillations, and inter-area mode oscillations are some of the different types of power system oscillations. The important advantage of increased energy transmission capability over the current interconnector is achieved through oscillation dampening. To reduce power system oscillations, several investigations have been carried out [54,55]. To

identify the optimal control input parameters of a unified power flow controller (UPFC) for damping power system oscillations, a comparative analysis with the direct component of torque (DCT), minimum singular value (MSV), Hankel singular value (HSV), and residue has been proposed [56]. A damping control system, which is based on a generalized power-incorporated current controller, is a third-generation FACTS device that is presented in [57], to investigate its effect on reducing low-frequency oscillation.

VII. CONCLUSIONS

The implementation of Flexible AC Transmission System (FACTS) devices, such as the UPFC, has emerged as a promising solution for improving the utilization of existing power in modern power systems. UPFC, with its ability to control active and reactive power flows, regulate bus voltage and current flow, and manage up to three transmission power system parameters simultaneously, has attracted significant attention in recent years. This paper has highlighted the integration of artificial intelligence (AI) techniques into the placement of UPFC, which can optimize energy management using machine learning and optimization algorithms. The proposed method ensures the optimal placement of UPFC, leading to enhanced power transfer capabilities and improved power quality, resulting in substantial cost savings. The use of AI in the placement of UPFC is a novel and exciting approach with great potential for the future of energy systems, and its scalability makes it a widely applicable solution. Overall, this technique can significantly enhance the performance and reliability of energy systems, which is crucial in today's world.

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